

# Clinical Risk Assessment Report: Patient 74

## 1. Primary Outcome & Risk Summary

- Target: Occult Lymph Node (OLN) Metastasis
- Model Prediction: Negative for Metastasis
- Prediction Confidence: Moderate Confidence
- Absolute Predicted Risk: 9.1% (Diagnostic Threshold: 22.1%)
- Relative Risk Multiplier: 0.41x the diagnostic threshold

### Clinical Takeaway:

The machine learning model classified this patient as Low-Risk for Occult Lymph Node (OLN) Metastasis. The primary driver determining this prediction is the Small Zone Emphasis (PET) (GLZLM\_SZE.PET), which reduced the patient's absolute risk by 2.0%. Biologically, this feature reflects the proportion of small, fine texture zones, often correlating with a highly chaotic and erratic micro-architecture.

## 2. Key Pathological & Imaging Drivers (Elevating Risk)

No major risk-elevating features observed in the top drivers.

## 3. Protective / Mitigating Factors (Reducing Risk)

**Small Zone Emphasis (PET) (GLZLM\_SZE.PET)** **Value: -1.73 [Bottom 6%] (-2.0%)**

*Reflects the proportion of small, fine texture zones, often correlating with a highly chaotic and erratic micro-architecture.*

**Patient Age (Age)** **Value: 69.0 [Avg (73th %ile)] (-1.1%)**

*Patient age influences immune surveillance, tissue microenvironment, and baseline metabolic rates.*

**Large Zone Emphasis (CT) (GLZLM\_LGZE.CT)** **Value: -0.1 [Avg (66th %ile)] (-1.0%)**

*Reflects the presence of large homogeneous areas; lower values suggest a chaotic, heterogeneous microenvironment.*

**Long Run High Gray-Level Emphasis (PET) (GLRLM\_LRHGE.PET)** **Value: -0.63 [Bottom 20%] (-0.9%)**

*Points to continuous, elongated bands of high-density or highly active tumor tissue.*

**Carcinoembryonic Antigen (CEA)** **Value: 1.5 [Bottom 15%] (-0.8%)**

*Elevated levels are strongly associated with increased tumor burden and potential for micrometastasis.*

**Tumor Sphericity (CT) (SHAPE\_Sphericity(onlyFor3DROI)).CT)** **Value: 0.62 [Avg (73th %ile)] (-0.7%)**

*Quantifies how closely the tumor resembles a perfect sphere; lower values indicate irregular, spiculated, or highly invasive margins.*

**75th Percentile Tumor Density (CONVENTIONAL\_HUQ3)** **Value: -1.46 [Bottom 10%] (-0.7%)**

*Reflects the overall physical density, solid components, and general attenuation characteristics of the tumor.*

**Tumor Local Contrast (PET) (GLCM\_Contrast(=Variance).PET)** **Value: -0.5 [Avg (37th %ile)] (-0.6%)**

*Measures local variations and sharp transitions in density or metabolism, signifying an unstructured and variable tumor matrix.*

Machine Learning Feature Attribution (SHAP Decision Path)

